

Species Distribution Modeling for Eastern Oyster Habitat

M. Fazio

2026-10-05

Table of contents

1	Introduction	1
2	Methods	8
2.1	Data Processing and Analysis	8
2.1.1	Software and Packages	8
2.1.2	Study Area	8
2.1.3	Water Quality Portal Data	8
2.1.4	Water Quality Data	9
2.2	Spatial interpolation	15
3	Results	15
4	Discussion	15
5	Conclusion	15
6	References	15

1 Introduction

The eastern oyster (*Crassostrea virginica*) are viewed as a foundational species as they are directly responsible for creating and providing numerous ecosystem goods and services such as food and habitat production, shoreline stabilization and protection, water quality improvement, nutrient cycling, and carbon sequestration (Geselbracht et al. 2024; Grabowski et al. 2012; Barnes et al. 2007). Additionally, they provide numerous economic benefits such as supporting oyster fisheries, recreational harvesting, tourism, along with the associated employment that

goes along with those industries (Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021). In one study, Grabowski et al. (2012) estimated the average annual value to range from \$10,325 to \$99,421 per hectare (2011 Dollars).

In recent decades, oyster reefs have experienced substantial declines, with only an estimated 15% remaining worldwide, and 50 to 80% being lost within the Gulf of (America) alone (Pollack et al. 2012; Peyre, Marshall, and Sable 2021). The decline is largely attributed to a combination of anthropogenic activities such as over harvesting, habitat alteration, sedimentation and water quality decline, disease and changing hydrology (Peyre, Marshall, and Sable 2021; Geselbracht et al. 2024). In response to these sharp declines, many restoration efforts have been undertaken with varying strategies and success rates; recently, planning documents released by the Deepwater Horizon Natural resource Damage Assessment Trustees called for large-scale development of reef networks to increase resilience (Geselbracht et al. 2024).

Oyster decline in the Pensacola Bay System

One such system to experience oyster decline is the Pensacola Bay System (PBS). The PBS is a large estuarine system located in Escambia and Santa Rosa counties in northwest Florida. The system is comprised of multiple waterbody's including Pensacola Bay, Escambia Bay, Blackwater Bay, East Bay, and Santa Rosa Sound; and is influenced heavily by freshwater input from the Escambia, Blackwater and Yellow Rivers (Christensen et al. 1997). Historically, the PBS supported extensive oyster reefs that provided critical habitat for a variety of species and substantial economic value to the surrounding communities (Christensen et al. 1997). However, over the past several decades, the extent of oyster habitat within the PBS has declined significantly due to a combination of factors including over harvesting, habitat degradation, and water quality issues. With some estimates indicating a decline of 15,000ha to just 91.5 ha of oyster habitat between the years of 1883 and 2021 (Geselbracht et al. 2024). This decline has prompted efforts to restore oyster reefs within the PBS (Pensacola and Perdido Bays Estuary Program n.d.; The Nature Conservancy 2021), with multiple studies aiming to identify the most suitable areas within the estuary to focus restoration efforts (Geselbracht et al. 2024; Christensen et al. 1997; Pensacola and Perdido Bays Estuary Program n.d.).

With limited resources available for restoration, it is critical to identify areas where chances of oyster survival and long-term reef establishment are at their highest, and where poor site selection could ultimately lead to unsatisfactory results. As with any ecosystem, the environmental conditions within the PBS vary greatly in time and space, and such variations can have great impacts on oyster survival and establishment. Therefore, it is important to understand the environmental factors that drive oyster occurrence so that scientist and policy makers can make informed decisions regarding said restoration site selection (Pollack et al. 2012; Barnes et al. 2007; Geselbracht et al. 2024).

Environmental controls on Eastern oyster habitat

Oyster recruitment, survival, and reef establishment are influenced by complex interactions between numerous physical, chemical, and biological environmental factors. As such, determining the most influential factors is a multivariate problem that requires careful consideration of each

factor and its interactions with others (Peyre, Marshall, and Sable 2021; Pollack et al. 2012; Barnes et al. 2007; Christensen et al. 1997). To help narrow the focus, the most commonly cited environmental factors were examined in the literature and are summarized here.

Perhaps the most influential and widely reported environmental driver of oyster suitability is salinity. Representing an important environmental gradient within estuarine systems that influences physiology, recruitment, disease, predation, and survival. Other important characteristics often reported can be seen in Table 1 and include: dissolved oxygen, temperature, depth, turbidity, and bottom type (substrate) (Peyre, Marshall, and Sable 2021; Pollack et al. 2012; Barnes et al. 2007; Geselbracht et al. 2024; Christensen et al. 1997).

Table 1: Common environmental drivers of Eastern Oyster habitat suitability.

Factor	Description	Sources
Salinity	Controls feeding, growth, reproduction, recruitment, disease, predation, and survival.	(Barnes et al. 2007; Birch et al. 2021; Christensen et al. 1997; Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021; Pollack et al. 2012)
Temperature	Regulates metabolism, filtration, growth, spawning, and larval development.	(Barnes et al. 2007; Christensen et al. 1997; Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021; Pollack et al. 2012)
Substrate type	Stable hard substrate supports larval attachment and reef persistence, whereas soft or rapidly accumulating sediments can bury or destabilize oysters.	(Barnes et al. 2007; Birch et al. 2021; Christensen et al. 1997; Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021)
Dissolved oxygen	Adequate oxygen supports respiration, feeding, and growth.	(Christensen et al. 1997; Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021; Pollack et al. 2012)
Hydrology / water movement	Water movement regulates salinity and transports larvae, oxygen, and food.	(Barnes et al. 2007; Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021)
Depth / bathymetry	Depth influences exposure, temperature, oxygen conditions, and restoration feasibility.	(Christensen et al. 1997; Geselbracht et al. 2024; Pollack et al. 2012)
Turbidity / sedimentation	High suspended-sediment loads can interfere with filtration and settlement. Sediment deposition can smother oyster reefs.	(Peyre, Marshall, and Sable 2021; Pollack et al. 2012)
Food availability / chlorophyll a	Chlorophyll a is often used as a proxy for food availability.	(Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021)

It should be recognized that not all environmental factors are adequately spatially distributed or represented within the available datasets, and therefore careful consideration should be given when selecting the final dataset used for modeling habitat suitability (Peyre, Marshall, and Sable 2021; Geselbracht et al. 2024; Christensen et al. 1997).

Environmentally Driven Habitat Modeling

Quantifying the relationships between the distribution of species and the species-environment interactions have a long history in ecological research, having been described as far back as the late 17th early 18th century in which researchers observed biological patterns in terms of their relationships with geographical and/or environmental gradients (Rushton, Ormerod, and Kerby 2004; Elith and Leathwick 2009). Since then, the propagation of large-scale spatial datasets, the development and integration of GIS, increased access to statistical packages and computational power have allowed for much more development in modeling complex species-environment interactions (Johnson and Gillingham 2005).

Habitat Suitability Indices (HSIs)

HSIs use expert-driven variable selection and classification methods that transform environmental characteristics into standardized estimates of habitat suitability, and have often been used as decision support tools in restoration frameworks due to their relatively low cost and flexibility (Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024).

Further, such models have already been used substantially in Eastern oyster restoration throughout the Gulf Coast region in multiple different systems showing their applicability across spatial scales (Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024; Christensen et al. 1997)

Within the PBS, HSIs have been used to develop GIS-based and estuary-specific HSMs that integrate biological, physical, chemical, and management-related variables in order to identify suitable restoration areas with the system. In the mid 1990's Christensen et al. (1997) developed a HSI for the eastern oyster in the PBS to demonstrate the model frameworks usefulness in illustrating species suitability in a system wide approach. Geselbracht et al. (2024) then followed and further expanded on the approach incorporating management factors into the model, adding a layer of regulatory influence to the model output versus the purely environmental analysis of the prior. Additionally, the Pensacola and Perdido Bay Estuary Program recently built on-top of Geselbracht et al. (2024) by incorporating some of their own data and refining some of the weighting (Pensacola and Perdido Bays Estuary Program n.d.).

These previous studies alone highlight the strengths and weakness of the HSI approaches. Some of the strengths include interoperability, adaptability, and the integration of ecological knowledge, but the results entirely rely on expert decisions regarding variable selection, suitability functions, weighting and scoring, and how to aggregate the results. All of these present opportunity for differing outputs even if the same inputs were used by different groups (Geselbracht et al. 2024; Johnson and Gillingham 2005).

Species Distribution Modeling

Species distribution models (SDMs) can be broadly defined as numerical tools that combine observations of species occurrence or abundance with environmental estimates (Elith and Leathwick 2009). When combined with Geographic Information Systems (GIS), these predictive models have the ability to visualize the distribution and infer the suitability of species habitats. Further, these maps allow conservation professionals to concentrate and target their efforts in the hope to make more effective decisions (Johnson and Gillingham 2005).

Species distribution models as an empirical alternative or complement

1. Introduce species distribution models (SDMs).
 - SDMs quantitatively relate known species distributions to environmental conditions.
 - They can be used for ecological explanation and spatial prediction.
 - Citation: (Elith and Leathwick 2009)
2. Different habitat-modeling approaches can produce different spatial predictions.
 - Model selection and environmental-variable selection can influence areas identified as suitable.
 - Comparing modeling approaches can reveal areas of agreement and disagreement.
 - Citation: (Johnson and Gillingham 2005)
3. Empirical SDMs provide an alternative method of estimating habitat relationships directly from observed species occurrence.
 - Establish the conceptual transition from HSI approaches to MaxEnt.
 - Citations: (Elith and Leathwick 2009; Johnson and Gillingham 2005)

4. Maximum Entropy species distribution modeling

1. Introduce MaxEnt as a presence-background species distribution modeling framework.
 - Particularly useful where reliable absence observations are unavailable.
 - Relates environmental conditions at known occurrence locations to environmental conditions available throughout the modeling region.
 - Citations: (Phillips, Anderson, and Schapire 2006; Elith et al. 2011)
2. Explain why MaxEnt is appropriate for Eastern oyster occurrence data in PBS.
 - FWC oyster-bed data provide known oyster occurrences without systematically sampled absence locations.
 - Presence-background modeling therefore fits the available data structure.
 - Citations: (Phillips, Anderson, and Schapire 2006; FWC-FWRI 2026)
3. Explain important considerations when implementing MaxEnt.
 - Background extent and sampling.
 - Spatial sampling bias.

- Predictor selection.
 - Regularization.
 - Feature classes and model complexity.
 - Citations: (Phillips et al. 2009; Merow, Smith, and Silander 2013; Elith et al. 2011)
4. Explain the need for model tuning and spatial validation.
 - MaxEnt should not be treated as an automated black-box model.
 - Model configuration should be evaluated systematically.
 - Spatially independent validation should be used where occurrence data exhibit spatial structure.
 - Citations: (Kass et al. 2021; Roberts et al. 2017; Valavi et al. 2019)

5. Research rationale, objectives, and hypotheses

1. Establish the primary research problem.
 - An estuary-specific HSI already exists for Eastern oyster restoration within PBS.
 - This study will independently develop an empirically fitted MaxEnt habitat-suitability model using observed oyster occurrences and environmental conditions.
 - Citations: (Geselbracht et al. 2024; Phillips, Anderson, and Schapire 2006)
2. Primary objective.
 - Develop a spatially explicit MaxEnt SDM representing relative Eastern oyster habitat suitability throughout PBS.
3. Secondary objective.
 - Quantitatively compare the MaxEnt suitability surface with the existing Pensacola Bay HSI.
4. Supporting objectives.
 - Identify environmental predictors associated with observed oyster distribution.
 - Evaluate MaxEnt predictive performance using spatially separated validation data.
 - Identify geographic areas where MaxEnt and the HSI agree or disagree.
 - Evaluate whether MaxEnt provides a reproducible empirical complement to HSI-based restoration planning.
 - Citations: (Johnson and Gillingham 2005; Geselbracht et al. 2024; Merow, Smith, and Silander 2013)
5. Research hypothesis.
 - The MaxEnt model is expected to identify patterns of habitat suitability consistent with established Eastern oyster habitat requirements.
 - Measurable agreement is expected between MaxEnt and the existing HSI, with spatial disagreement arising from differences between empirically fitted environmental relationships and predefined HSI suitability relationships.

- Citations: (Johnson and Gillingham 2005; Geselbracht et al. 2024)

2 Methods

2.1 Data Processing and Analysis

2.1.1 Software and Packages

All data processing, analysis, and modeling will be conducted using free open source software and packages to maximize accessibility and reproducibility. The majority of data manipulation and processing will be performed in the R programming language using the tidyverse suite of packages (Wickham et al. 2019; R Core Team 2024). Any data operations that need to be “spatially aware” will be performed using the Simple Features package (Pebesma 2018).

Analysis will be scripted and ran in Positron, an open-source integrated development environment developed by Posit that supports R, Python, scripts, notebooks, integrated consoles, along with interactive plots and data viewers.

2.1.2 Study Area

The study area for this project is the Pensacola Bay System which includes portions of Pensacola Bay, Escambia Bay, East Bay and Blackwater Bay. Boundaries of this study area were achieved by accessing Florida Department of Protections (FDEP) Waterbod IDs feature layer hosted on ArcGIS Online (FDEP 2005). ArcGIS allows direct access to hosted feature services via several formats. To avoid any potential compatibility issues, the GeoJSON format was chosen as this format is non-proprietary. Once retrieved, the dataset was filtered to include only relevant WBIDs (WBID = 548) and then combined to create a singular study area polygon. Creating the study area polygon is an important first step as it will be needed to extract relevant environmental data from much larger datasets in the coming data gathering.

2.1.3 Water Quality Portal Data

Water quality data is widely available for the entire United States via the EPA’s Water Quality Portal (WQP) which allows for a wide range of data querying for any data as it relates to Water Quality Monitoring programs (USGS (2026)). For this analysis, site data for Escambia and Santa Rosa County was queried using the WQP API then the study area polygon was used to extract only the relevant sites:

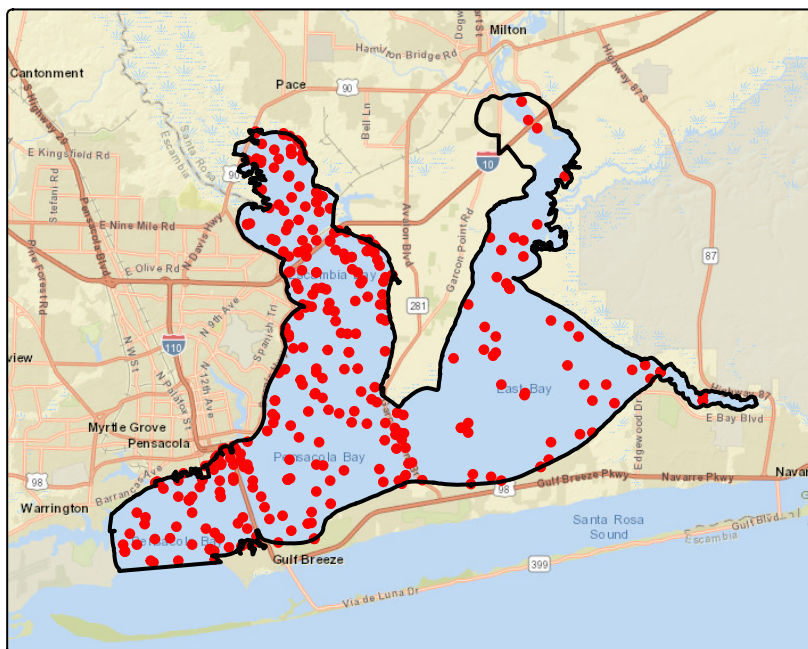


Figure 1: Pensacola Bay System with monitoring sites

2.1.4 Water Quality Data

Water quality data was selected based off of availability and relevance within literature. Each parameter was filtered to include just those measurements

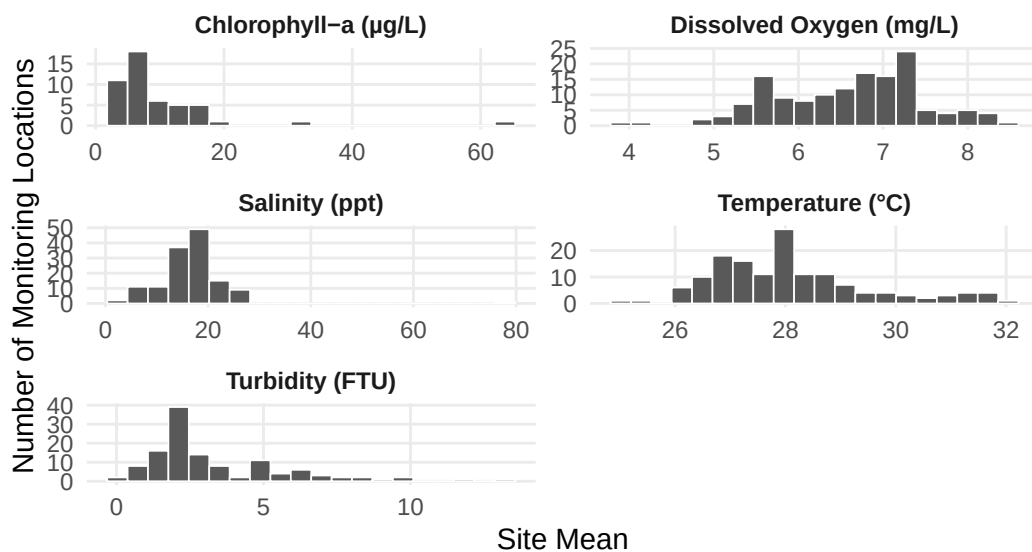
Table 2: Preliminary Environmental Variables Considered for the Eastern Oyster SDM

Variable	Obtained? (Y/N)	Data Type	Total Records	Data Source
Salinity	Y	Points	6994	National Water Quality Monitoring Council (2026)
Water temperature	Y	Points	7161	USGS (2026)
Substrate / bottom type	Y	Raster	NA	(ppbep?)
Water depth / bathymetry	N	Raster	National Centers for Environmental Information (2018)	NA

Variable	Obtained? (Y/N)	Data Type	Total Records	Data Source
Dissolved oxygen	Y	Points	6261	USGS (2026)
Chlorophyll-a	Y	Points	1847	USGS (2026)
Oyster bed locations	NA	NA	FWC-FWRI (2026)	NA

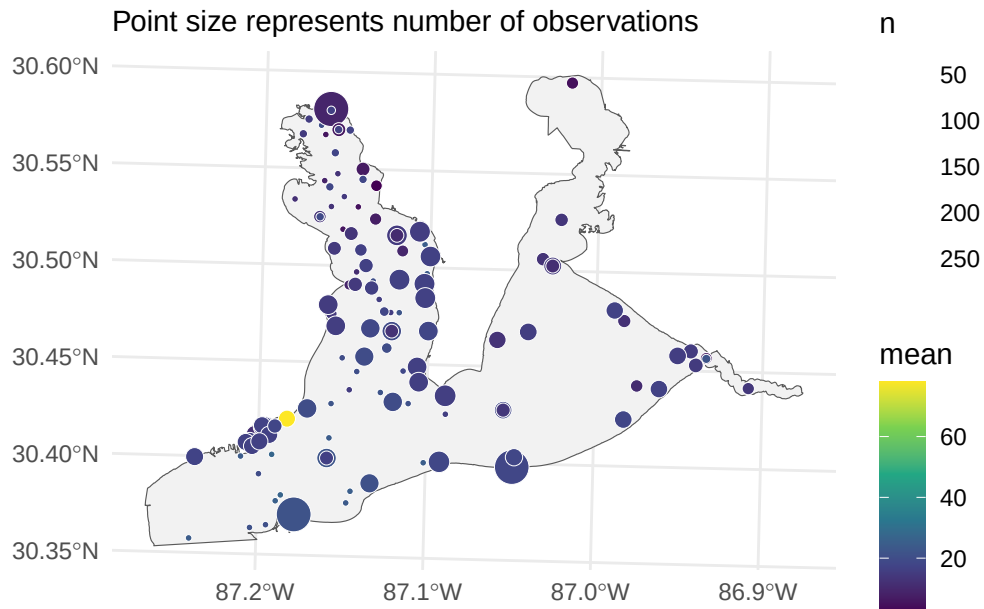
Distribution of Warm-Season Water Quality Conditions

Site-level means for monitoring locations in the Pensacola Bay System



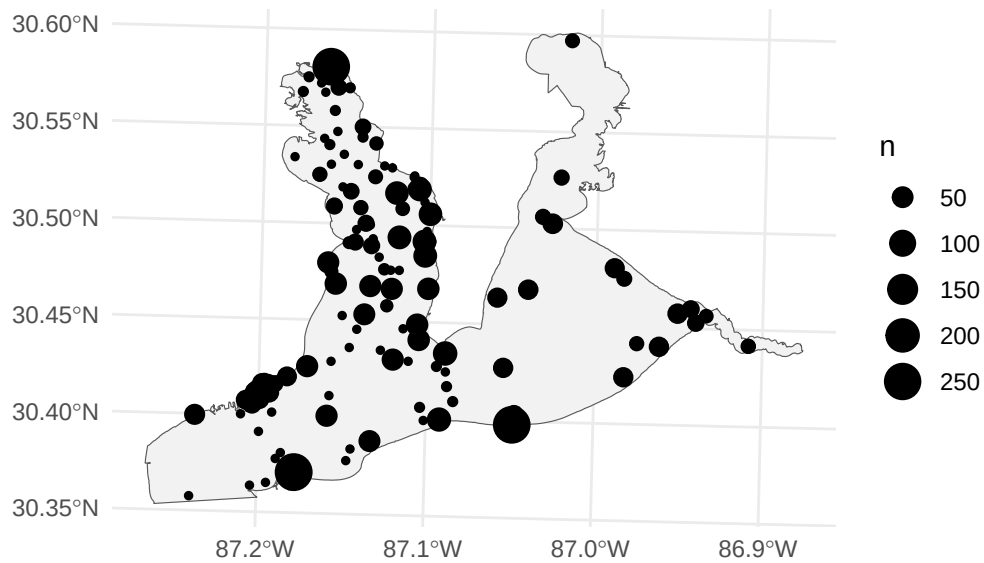
Warm-Season Salinity Monitoring Locations

Point size represents number of observations



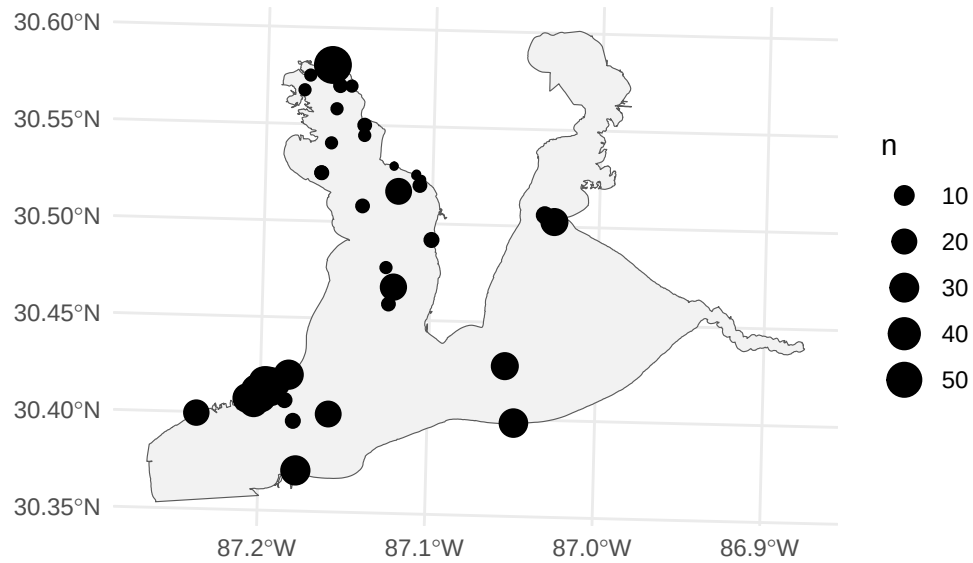
Warm-Season Temperature Monitoring Locations

Point size represents number of observations



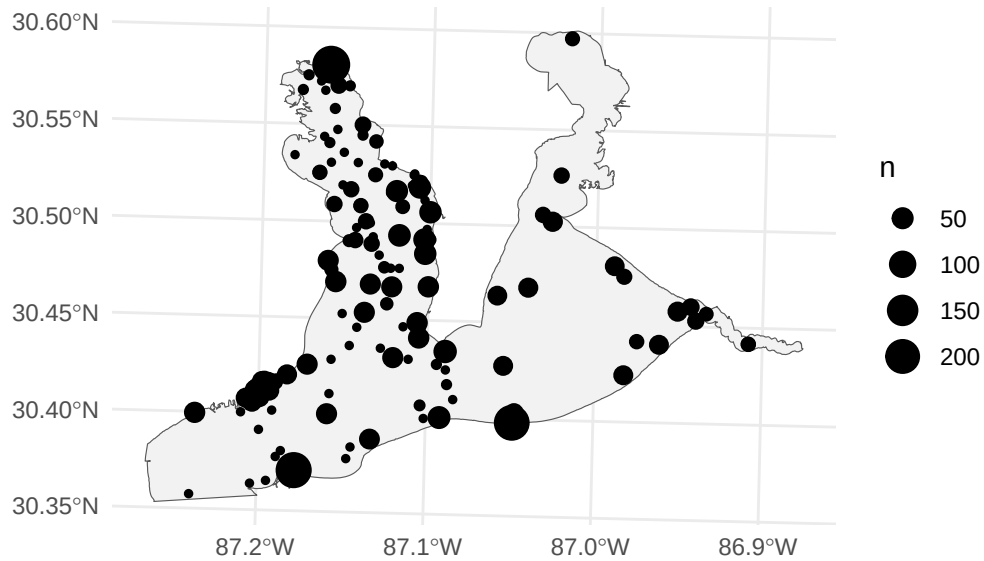
Warm-Season Chlorophyll-a Monitoring Locations

Point size represents number of observations



Warm-Season Dissolved Oxygen Monitoring Locations

Point size represents number of observations



Warm-Season Turbidity Monitoring Locations

Point size represents number of observations

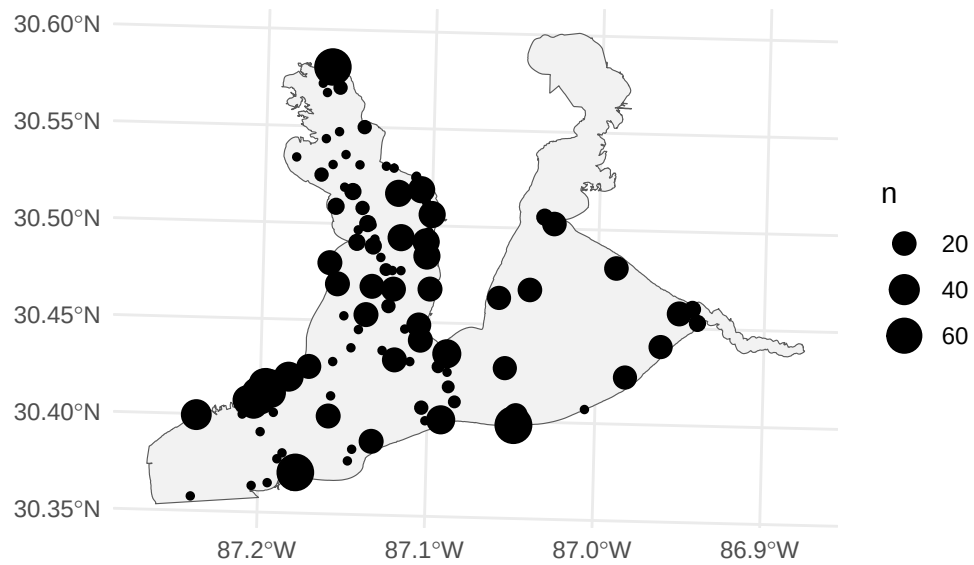


Table 3: Summary Statistics for Warm-Season Water Quality Data

Parameter	Sta- tions	Observa- tions	Mean	SD	Min	Q1	Me- dian	Q3	Max	First Sample	Last Sample
Chlorophyll-a (µg/L)	48	754	9.74	9.70	3.39	5.14	6.21	11.79	63.80	2005-05-03	2024-10-17
Dissolved Oxygen (mg/L)	145	3072	6.59	0.87	3.90	5.82	6.75	7.21	8.48	2005-05-03	2024-10-17
Salinity (ppt)	135	3401	16.55	7.35	2.31	14.02	16.54	18.91	77.85	2005-05-03	2024-10-17
Temperature (°C)	145	3442	28.11	1.43	25.01	27.05	27.92	28.74	32.04	2005-05-03	2024-10-17
Turbidity (FTU)	122	1563	3.33	2.41	0.03	1.85	2.35	4.65	13.22	2005-05-10	2024-10-17

2.2 Spatial interpolation

Points are interpolated into prediction rasters using inverse path distance weighting (IPDW) (package `idpw`). Blah blah blah blah

3 Results

4 Discussion

5 Conclusion

In progress...

6 References

- Barnes, T. K., A. K. Volety, K. Chartier, F. J. Mazzotti, and L. Pearlstine. 2007. "A Habitat Suitability Index Model for the Eastern Oyster (*Crassostrea virginica*), a Tool for Restoration of the Caloosahatchee Estuary, Florida." *Journal of Shellfish Research* 26 (4): 949–59. [https://doi.org/10.2983/0730-8000\(2007\)26%5B949:AHSIMF%5D2.0.CO;2](https://doi.org/10.2983/0730-8000(2007)26%5B949:AHSIMF%5D2.0.CO;2).
- Birch, A., R. Brumbaugh, B. DeAngelis, L. Geselbracht, A. Graves, J. Blair, and R. Jones. 2021. "Oyster Fisheries and Habitat Management Plan for the Pensacola Bay System." The Nature Conservancy. https://www.nature.org/content/dam/tnc/nature/en/documents/PBS_FloridaOyster-Fisheries-Habitat-Mgt-Plan_18May2021_Final.pdf.
- Christensen, J D, T A Battista, M E Monaco, and C J Klein. 1997. "Habitat Suitability Index Modeling and GIS Technology to Support Habitat Management: Pensacola Bay, Florida Case Study." U.S. National Oceanic; Atmospheric Administration. <https://repository.library.noaa.gov/view/noaa/55683>.
- Elith, Jane, and John R. Leathwick. 2009. "Species Distribution Models: Ecological Explanation and Prediction Across Space and Time." *Annual Review of Ecology, Evolution, and Systematics* 40 (December): 677–97. <https://doi.org/10.1146/annurev.ecolsys.110308.120159>.
- Elith, Jane, Steven J. Phillips, Trevor Hastie, Miroslav Dudík, Yung En Chee, and Colin J. Yates. 2011. "A Statistical Explanation of MaxEnt for Ecologists." *Diversity and Distributions* 17 (January): 43–57. <https://doi.org/10.1111/j.1472-4642.2010.00725.x>.
- FDEP. 2005. "Waterbody IDs (WBIDs)." Geospatial Open Data; Florida Department of Environmental Protection. <https://geodata.dep.state.fl.us/datasets/FDEP::waterbody-ids-wbids/about>.
- FWC-FWRI. 2026. "Oyster Beds in Florida." Geospatial Open Data; Florida Fish; Wildlife Conservation Commission, Fish; Wildlife Research Institute. <https://geodata.myfwc.com/datasets/myfwc::oyster-beds-in-florida/about>.

- Geselbracht, L., M. Johnston, B. M. DeAngelis, and A. Birch. 2024. “Estuary-Specific and Adaptive Habitat Suitability Index Model for the Eastern Oyster *Crassostrea Virginica* in the Pensacola Bay System, Florida, USA.” *Coastal Management* 52: 17–34. <https://doi.org/10.1080/08920753.2024.2335129>.
- Grabowski, Jonathan H., Robert D. Brumbaugh, Robert F. Conrad, Andrew G. Keeler, James J. Opaluch, Charles H. Peterson, Michael F. Piehler, Sean P. Powers, and Ashley R. Smyth. 2012. “Economic Valuation of Ecosystem Services Provided by Oyster Reefs.” *BioScience* 62 (October): 900–909. <https://doi.org/10.1525/bio.2012.62.10.10>.
- Johnson, Chris J., and Michael P. Gillingham. 2005. “An Evaluation of Mapped Species Distribution Models Used for Conservation Planning.” *Environmental Conservation* 32 (June): 117–28. <https://doi.org/10.1017/S0376892905002171>.
- Kass, Jamie M., Robert Muscarella, Peter J. Galante, Corentin L. Bohl, Gonzalo E. Pinilla-Buitrago, Robert A. Boria, Mariano Soley-Guardia, and Robert P. Anderson. 2021. “ENMeval 2.0: Redesigned for Customizable and Reproducible Modeling of Species’ Niches and Distributions.” *Methods in Ecology and Evolution* 12 (September): 1602–8. <https://doi.org/10.1111/2041-210X.13628>.
- Merow, Cory, Matthew J. Smith, and John A. Silander. 2013. “A Practical Guide to MaxEnt for Modeling Species’ Distributions: What It Does, and Why Inputs and Settings Matter.” *Ecography* 36 (October): 1058–69. <https://doi.org/10.1111/j.1600-0587.2013.07872.x>.
- National Centers for Environmental Information. 2018. “NOAA NOS Estuarine Bathymetry - Pensacola Bay (G130).” <https://doi.org/10.7289/V5XK8CWK>.
- National Water Quality Monitoring Council. 2026. *Water Quality Portal*. <https://doi.org/10.5066/P9QRKUVJ>.
- Pebesma, Edzer. 2018. “Simple Features for R: Standardized Support for Spatial Vector Data.” *The R Journal* 10 (1): 439–46. <https://doi.org/10.32614/RJ-2018-009>.
- Pensacola and Perdido Bays Estuary Program. n.d. “Pensacola Bay Oyster Restoration Initiative.” <https://www.ppbep.org/what-we-do/restoration/pensacola-bay-oyster-restoration-initiative>.
- Peyre, Megan K. La, Danielle A. Marshall, and Shaye E. Sable. 2021. “Oyster Model Inventory: Identifying Critical Data and Modeling Approaches to Support Restoration of Oyster Reefs in Coastal u.s. Gulf of Mexico Waters.” *Diversity and Distributions*. Vol. 17. U.S. Geological Survey. <https://doi.org/10.3133/ofr20211063>.
- Phillips, Steven J., Robert P. Anderson, and Robert E. Schapire. 2006. “Maximum Entropy Modeling of Species Geographic Distributions.” *Ecological Modelling* 190 (January): 231–59. <https://doi.org/10.1016/j.ecolmodel.2005.03.026>.
- Phillips, Steven J., Miroslav Dudík, Jane Elith, Catherine H. Graham, Anthony Lehmann, John Leathwick, and Simon Ferrier. 2009. “Sample Selection Bias and Presence-Only Distribution Models: Implications for Background and Pseudo-Absence Data.” *Ecological Applications* 19 (January): 181–97. <https://doi.org/10.1890/07-2153.1>.
- Pollack, Jennifer Beseres, Andrew Cleveland, Terence A. Palmer, Anthony S. Reisinger, and Paul A. Montagna. 2012. “A Restoration Suitability Index Model for the Eastern Oyster (*Crassostrea Virginica*) in the Mission-Aransas Estuary, TX, USA.” *PLoS ONE* 7 (July). <https://doi.org/10.1371/journal.pone.0040839>.

- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Roberts, David R., Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, et al. 2017. “Cross-Validation Strategies for Data with Temporal, Spatial, Hierarchical, or Phylogenetic Structure.” *Ecography*. Blackwell Publishing Ltd. <https://doi.org/10.1111/ecog.02881>.
- Rushton, S. P., S. J. Ormerod, and G. Kerby. 2004. “New Paradigms for Modelling Species Distributions?” *Journal of Applied Ecology*. <https://doi.org/10.1111/j.0021-8901.2004.00903.x>.
- The Nature Conservancy. 2021. “Pensacola East Bay Oyster Habitat Restoration Project.” The Nature Conservancy. <https://www.nature.org/content/dam/tnc/nature/en/documents/PEB-Oyster-Habitat-Restoration-Fact-Sheet-FINAL-7-13-21.pdf>.
- USGS, USDA, USEPA. 2026. “Water Quality Portal.” U.S. Geological Survey; U.S. Environmental Protection Agency; U.S. Department of Agriculture. <https://www.waterqualitydata.us/>.
- Valavi, Roozbeh, Jane Elith, José J. Lahoz-Monfort, and Gurutzeta Guillera-Arroita. 2019. “blockCV: An r Package for Generating Spatially or Environmentally Separated Folds for k-Fold Cross-Validation of Species Distribution Models.” *Methods in Ecology and Evolution* 10 (February): 225–32. <https://doi.org/10.1111/2041-210X.13107>.
- Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D’Agostino McGowan, Romain François, Garrett Grolemund, et al. 2019. “Welcome to the tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.